

Statistical Word Segmentation: Emergent Structure in a Next-Character RNN

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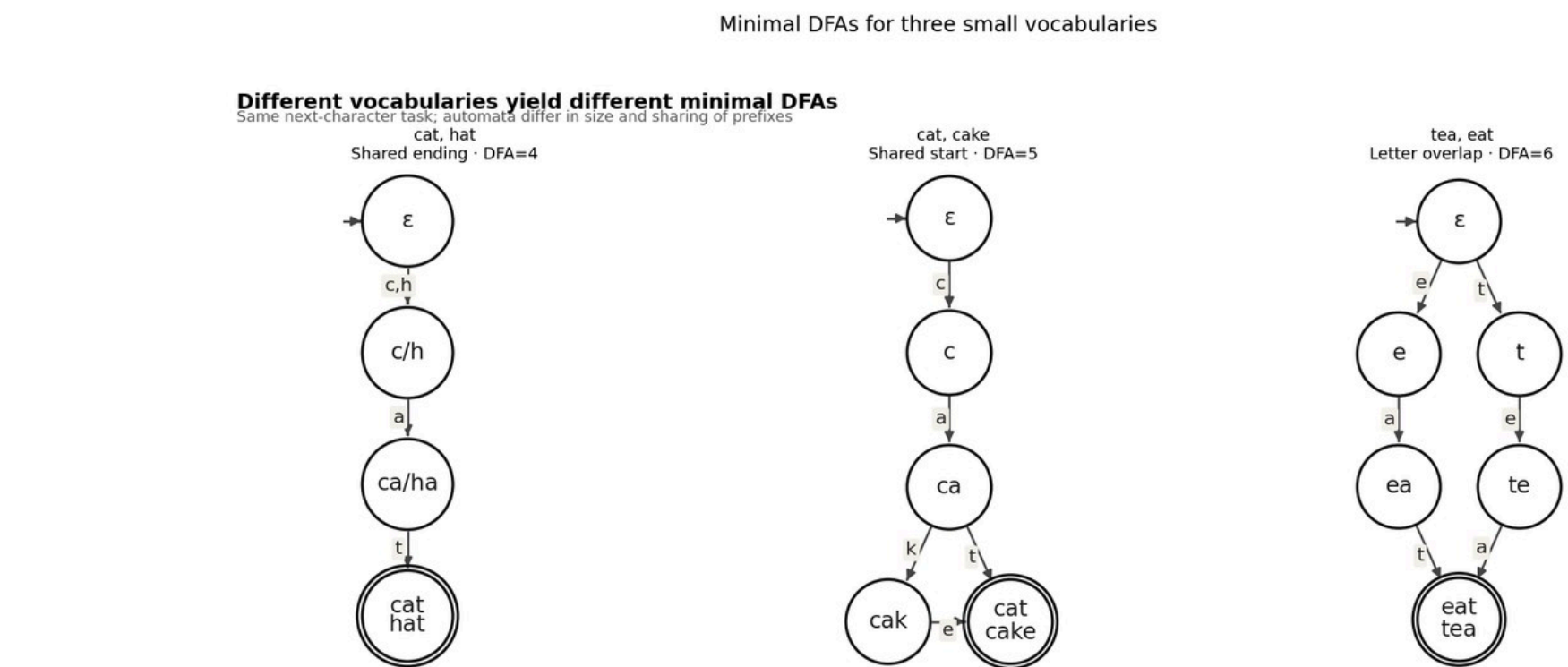


Code & figures on GitHub

Abstract

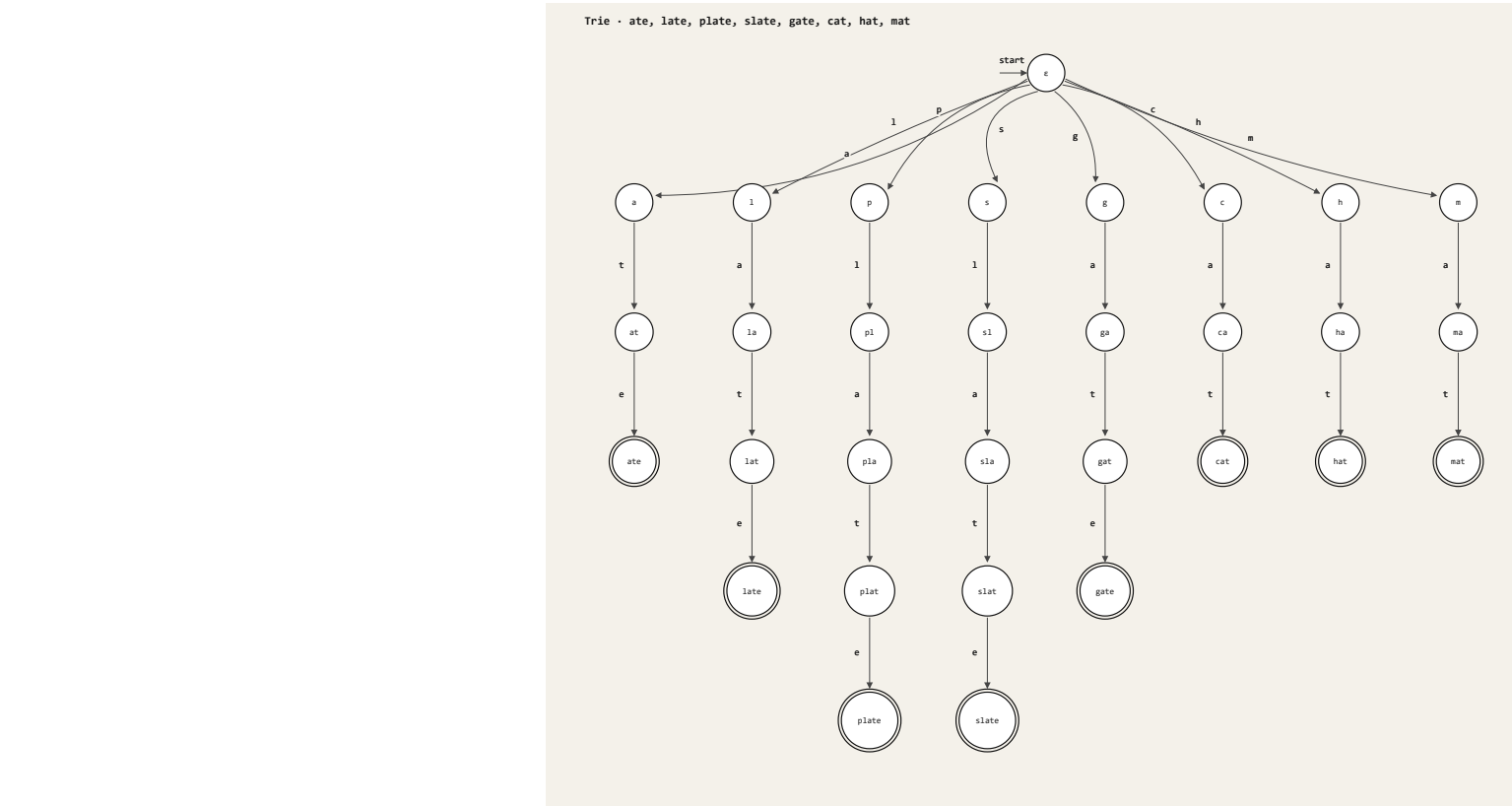
Eight-month-old infants can segment continuous speech by tracking transitional probabilities (Saffran, Aslin & Newport, 1996). We ask whether a vanilla Elman RNN trained only on next-character prediction developments aligned with word structure. After learning, the network generates legal vocabulary items, and its hidden states become a continuous embedding of the vocabulary's minimal DFA. On an eight-word *-ate/-at* demo, DFA state explains $\eta^2 \approx 0.88$ of condensed hidden variance; word identity is much weaker ($\eta^2 \approx 0.06$). Fifty mixed-length English vocab runs ($H=100$) show that dimensionality, training cost, and readouts track minimized DFA size. Weight matrices become letter-columnar in W_{ab} and locally clumped in W_{ba} , most clearly for small automata.

Fig 1 • Minimal DFAs



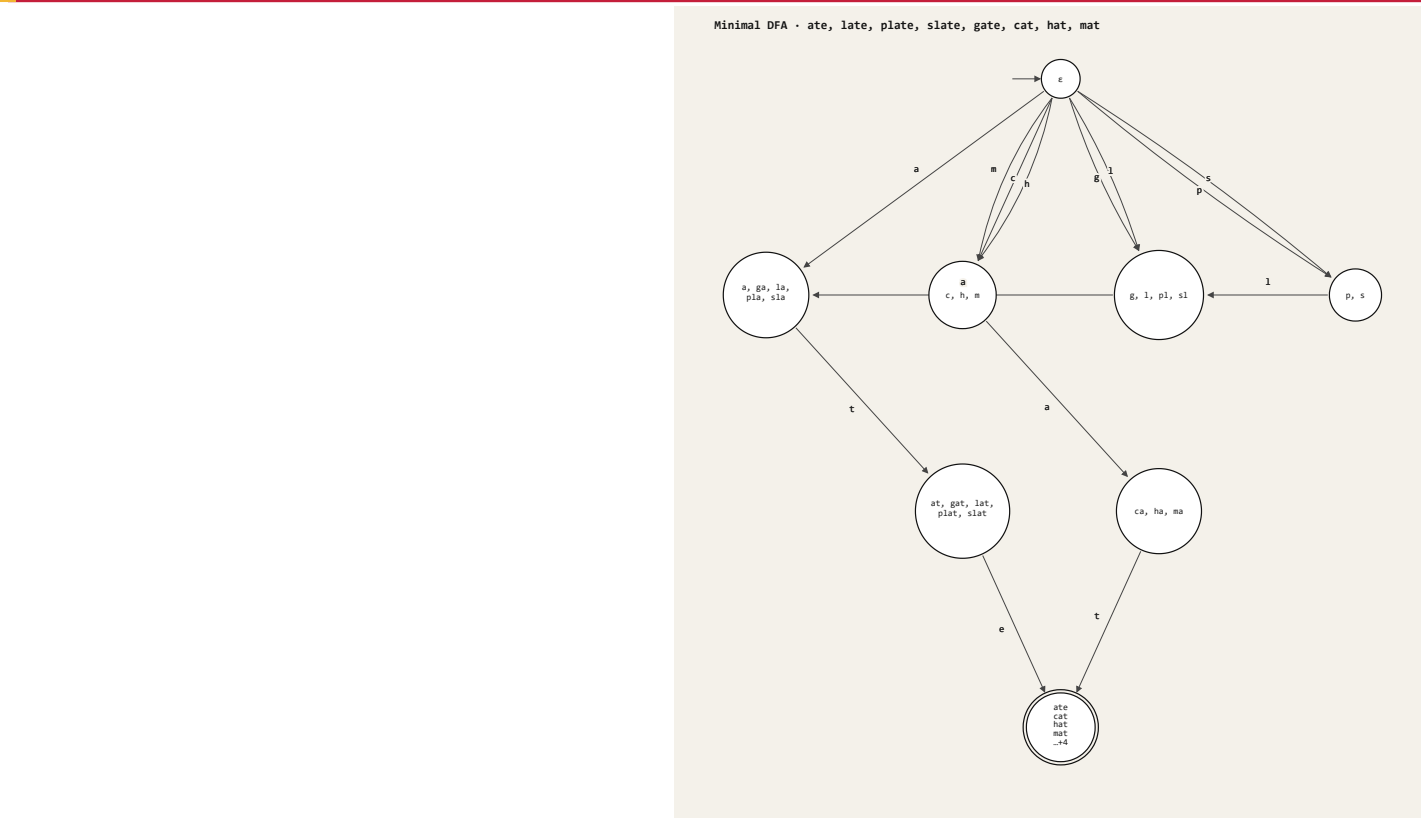
Same next-character task; automata differ in size and prefix sharing.

Fig 2 • Demo trie



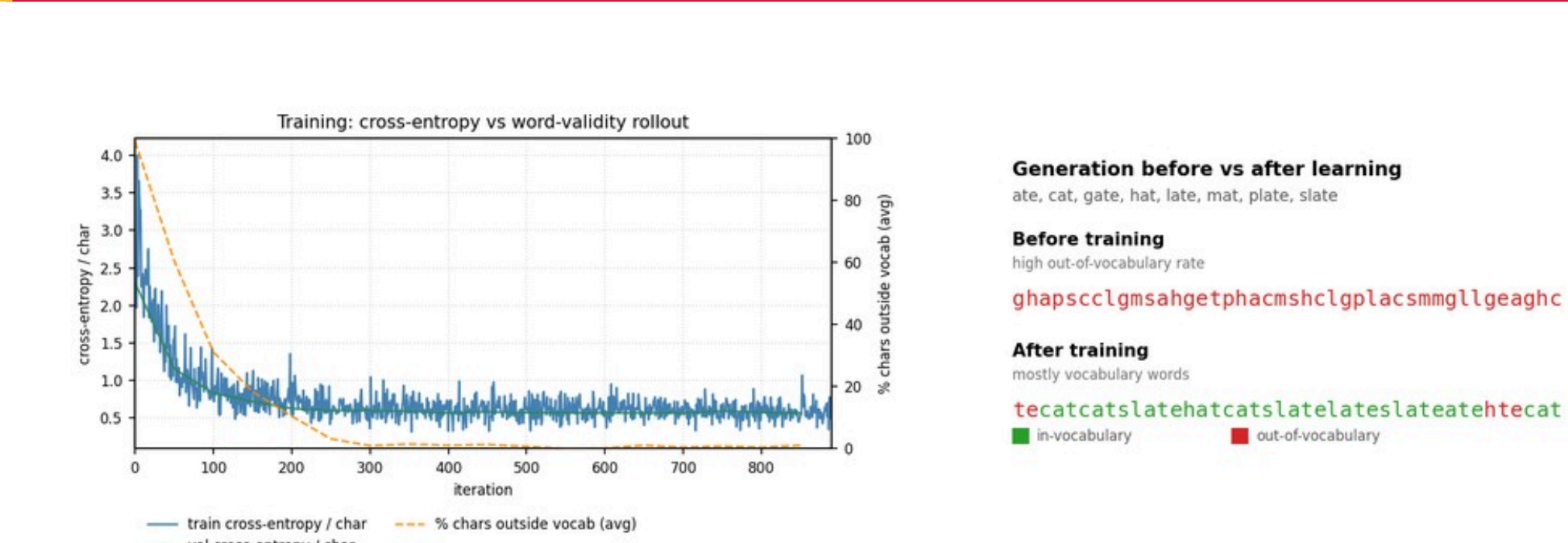
Prefix tree for *ate, late, plate, slate, gate, cat, hat, mat*.

Fig 2 • Demo minimal DFA



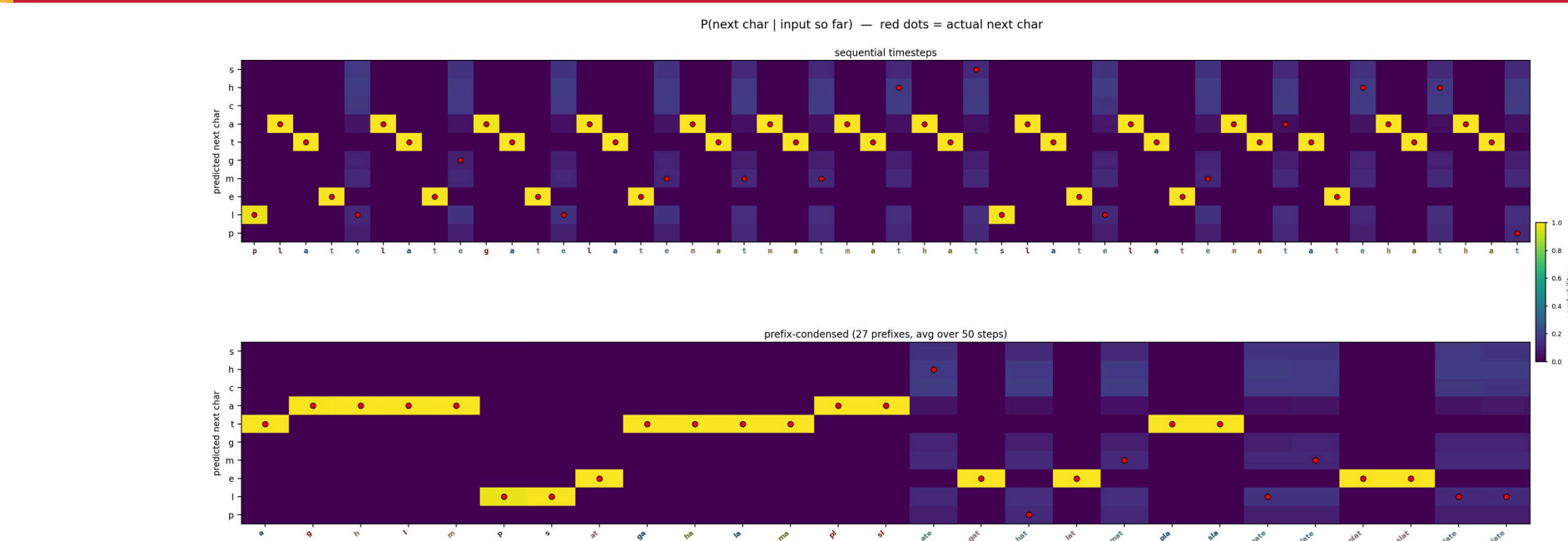
Shared states merge prefixes with identical futures (e.g. *a*g*a*/*a*l*a*/*a*).

Fig 3 • Learning & generation



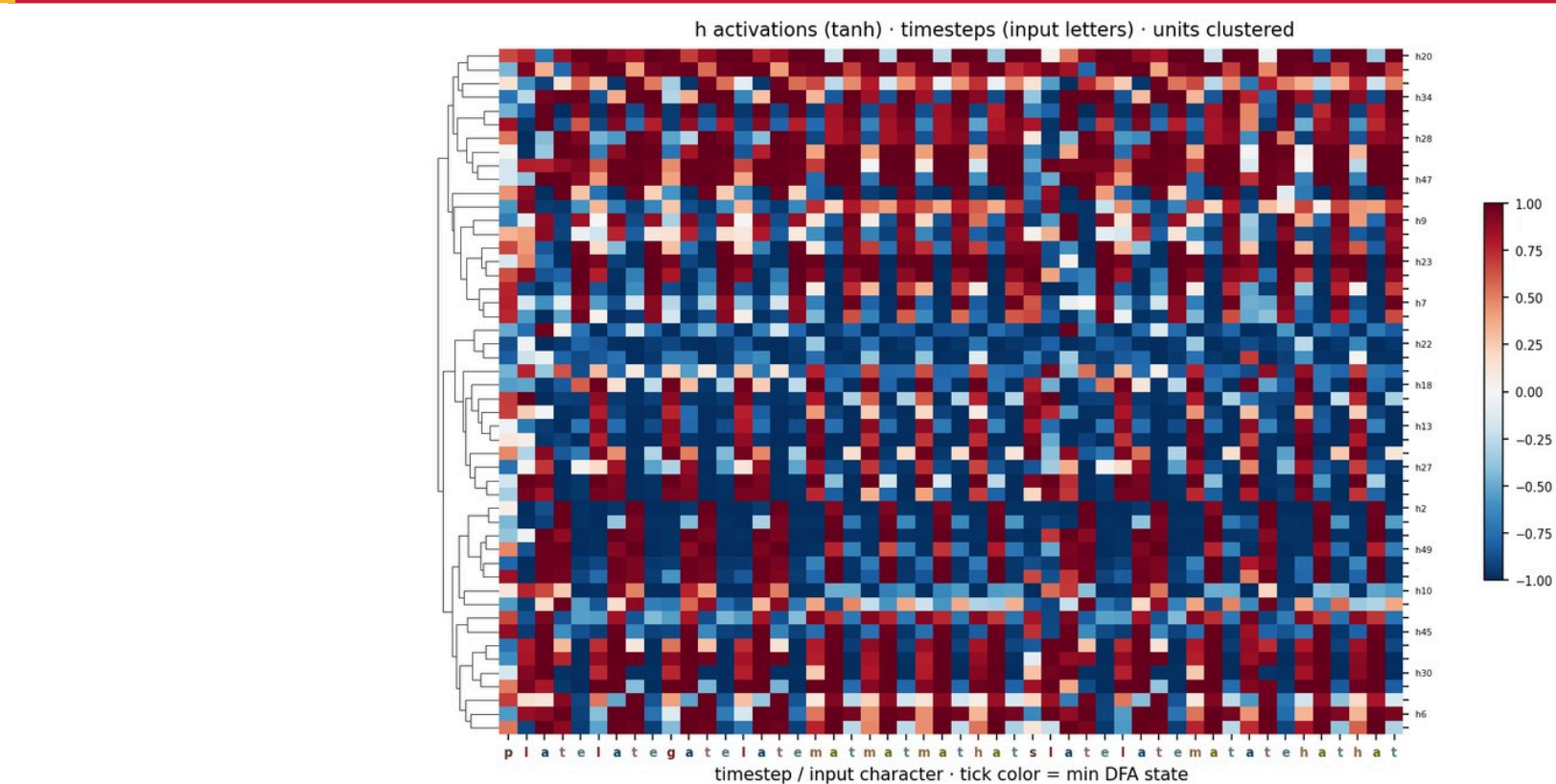
Loss and OOV collapse together; rollouts become legal vocabulary words.

Fig 4 • Next-character probabilities



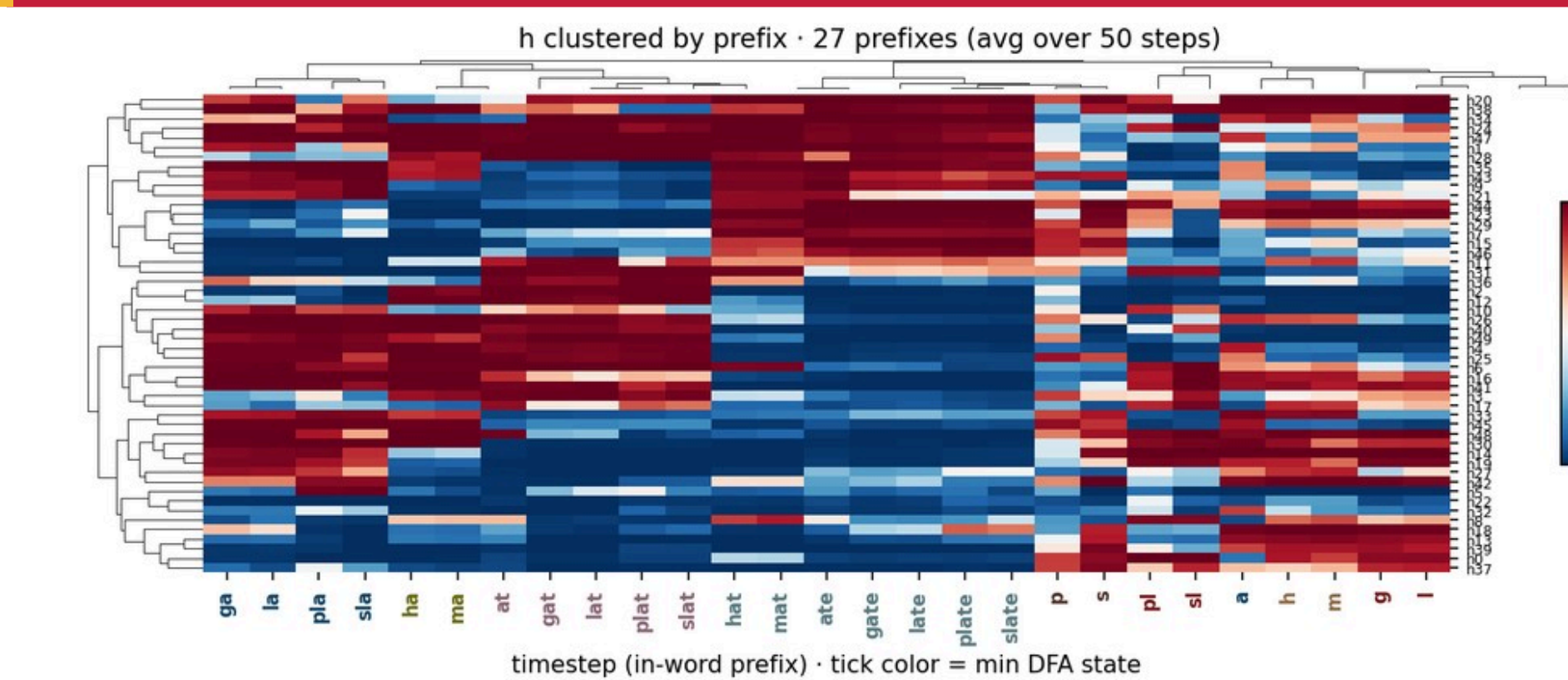
Softmax $P(\text{next char} | \text{input so far})$. Red dots = actual next character.

Fig 5 • Activation heatmap



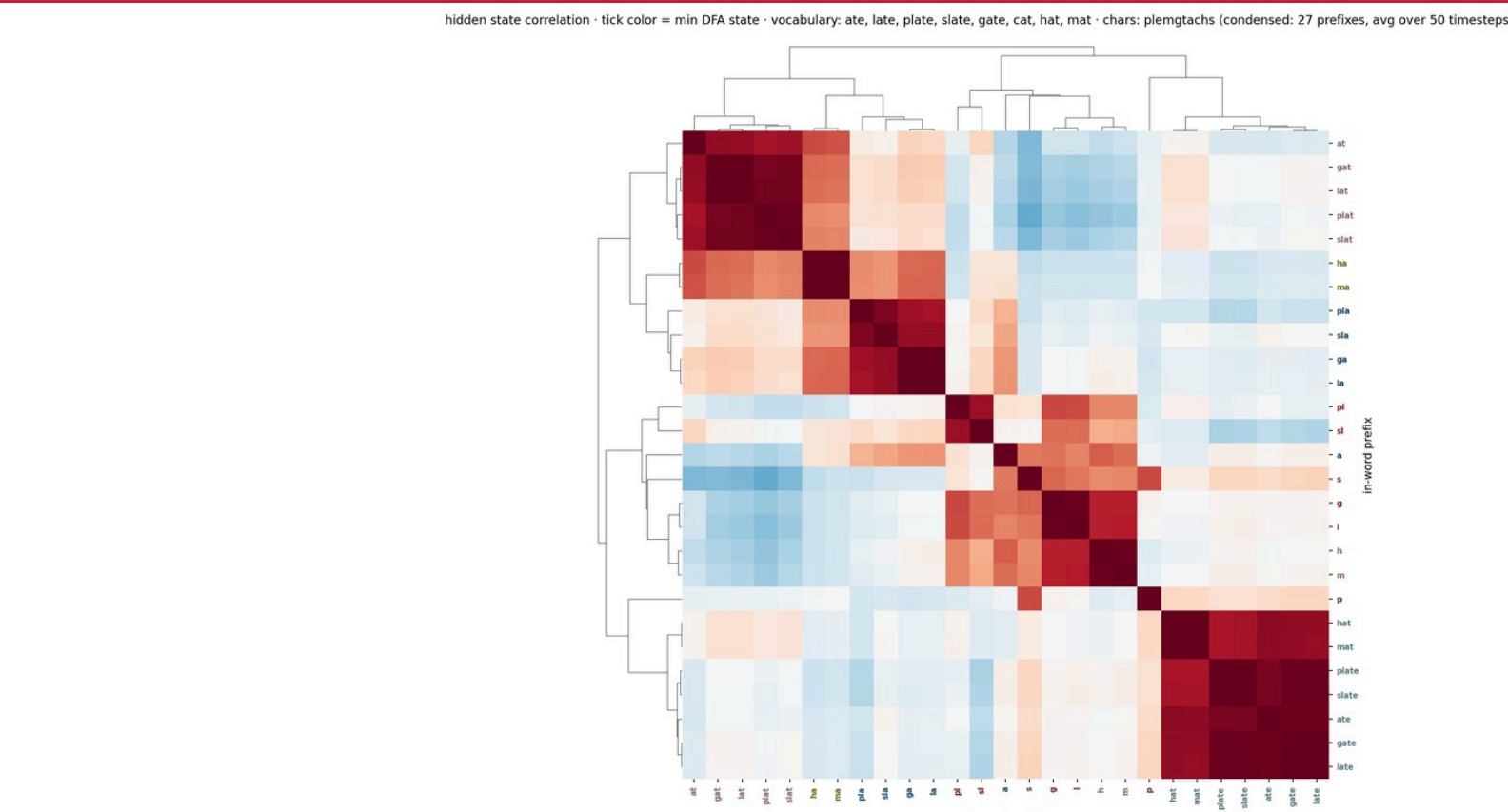
Hidden units over timesteps (input letters; units clustered).

Fig 6 • Activations by prefix



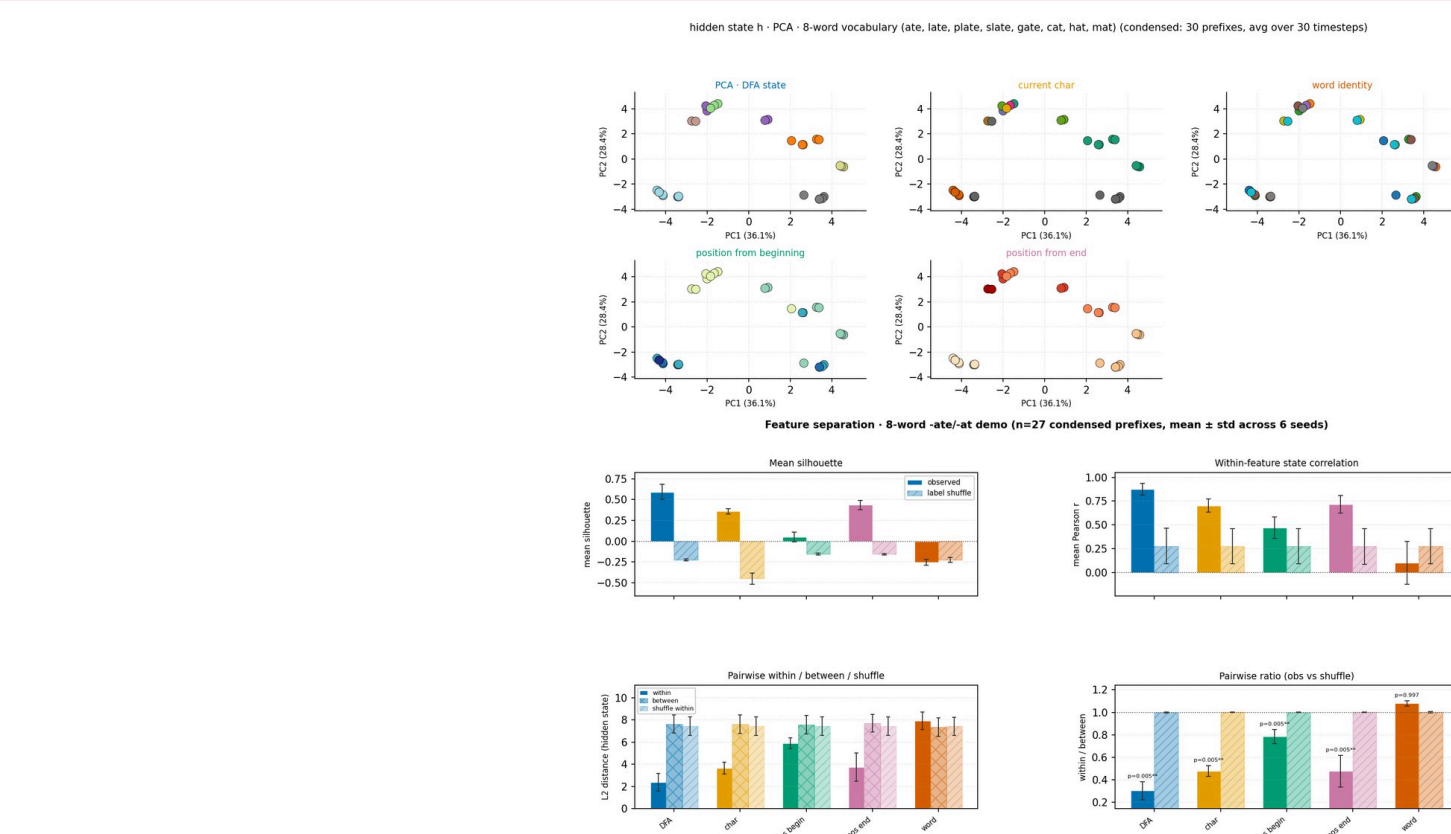
Activations clustered by in-word prefix. Tick color = DFA state.

Fig 7 • Hidden-state correlation



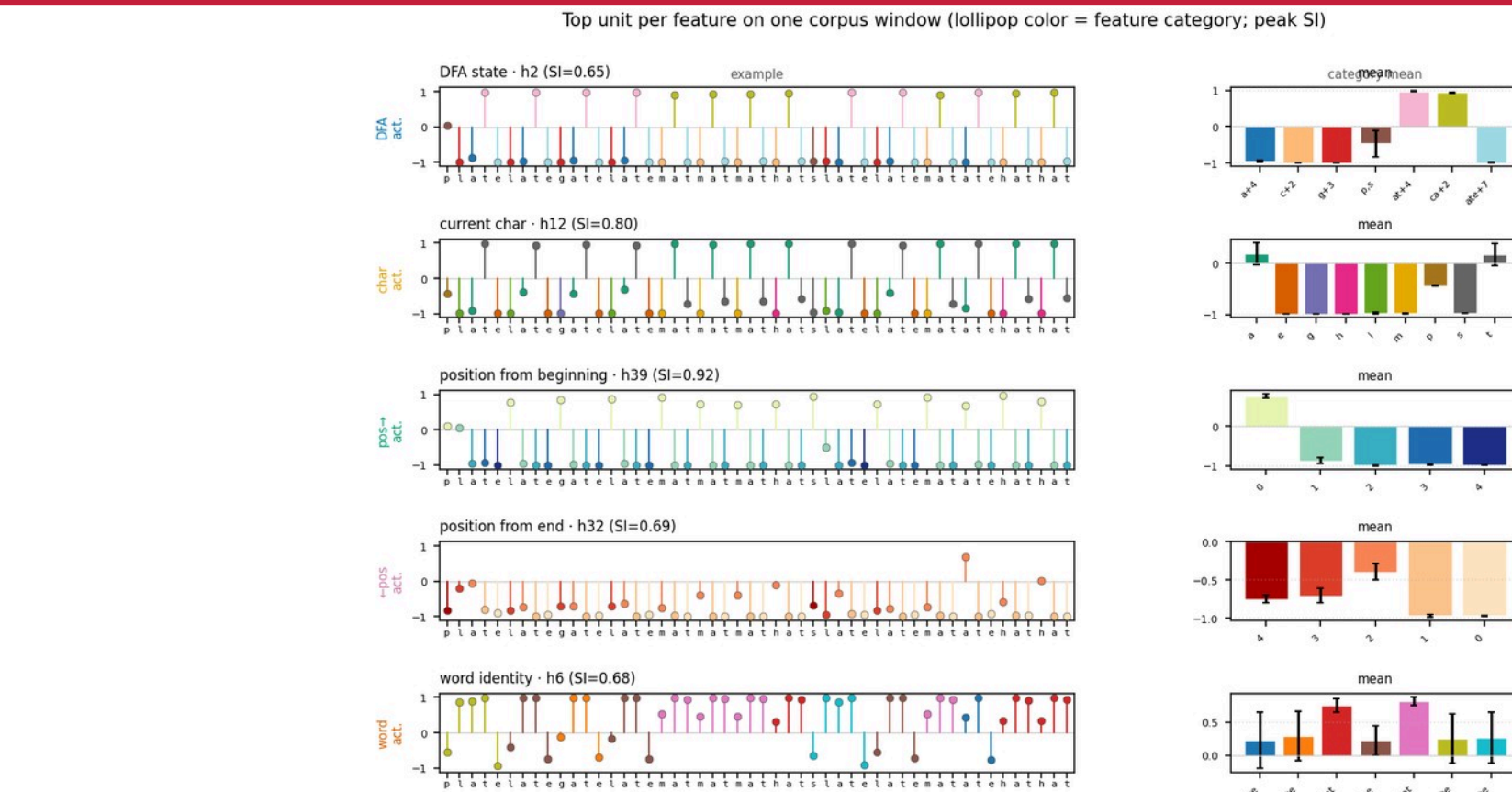
Prefix $\times$ prefix correlation. Blocks merge prefixes that share a future.

Fig 8 • DFA geometry & separation



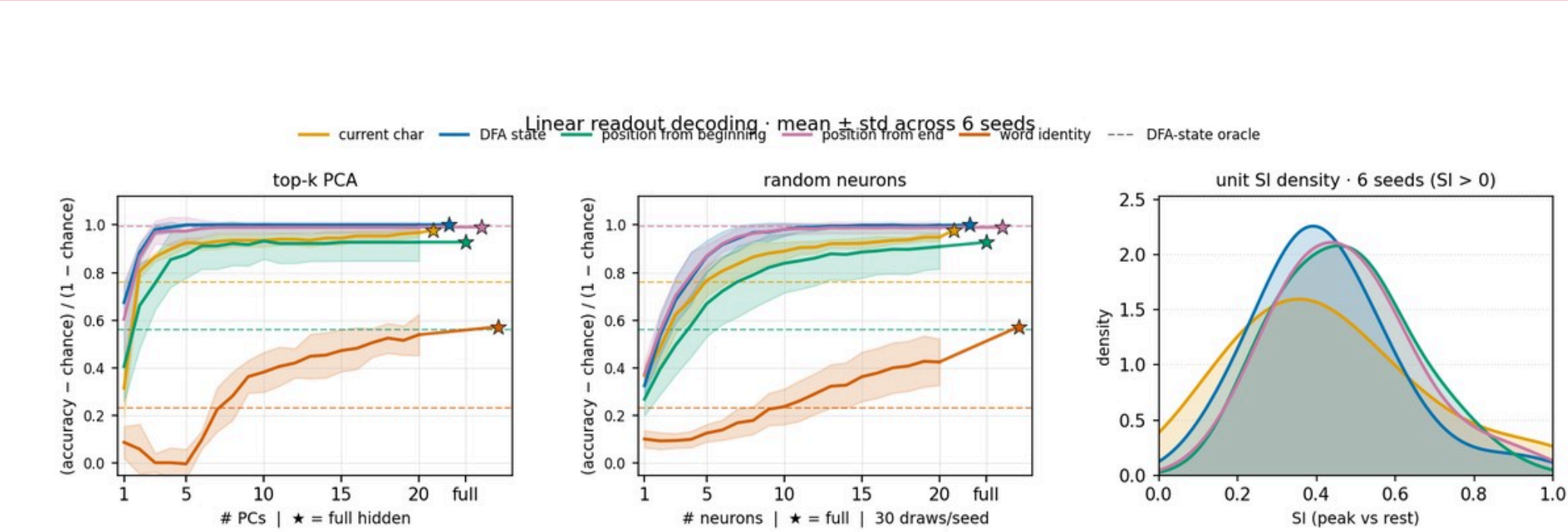
PCA of h by feature. DFA $\eta^2 \approx 0.88$; word identity ≈ 0.06 .

Fig 9 • Example selective units



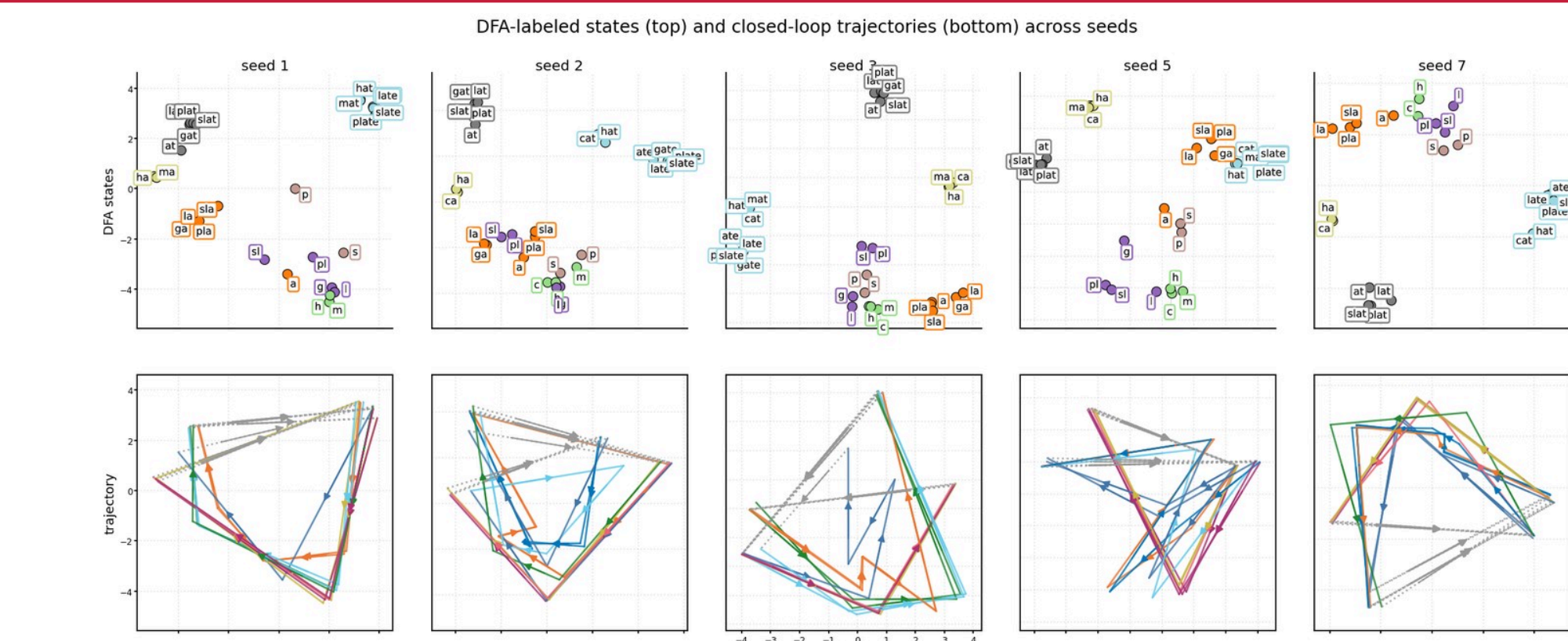
One top unit per feature on a shared corpus window.

Fig 10 • Linear decoding



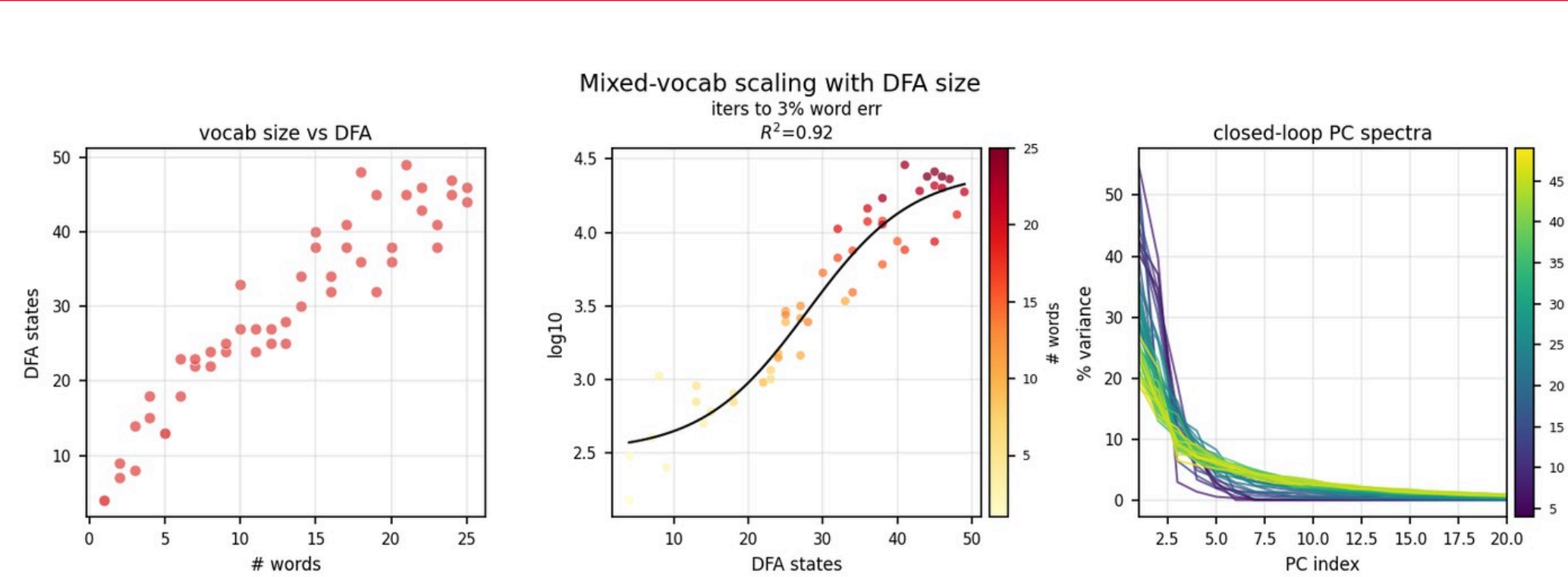
Mean $\pm$ std, 6 seeds. DFA/character saturate in a few PCs; word identity needs many more.

Fig 11 • Word trajectories



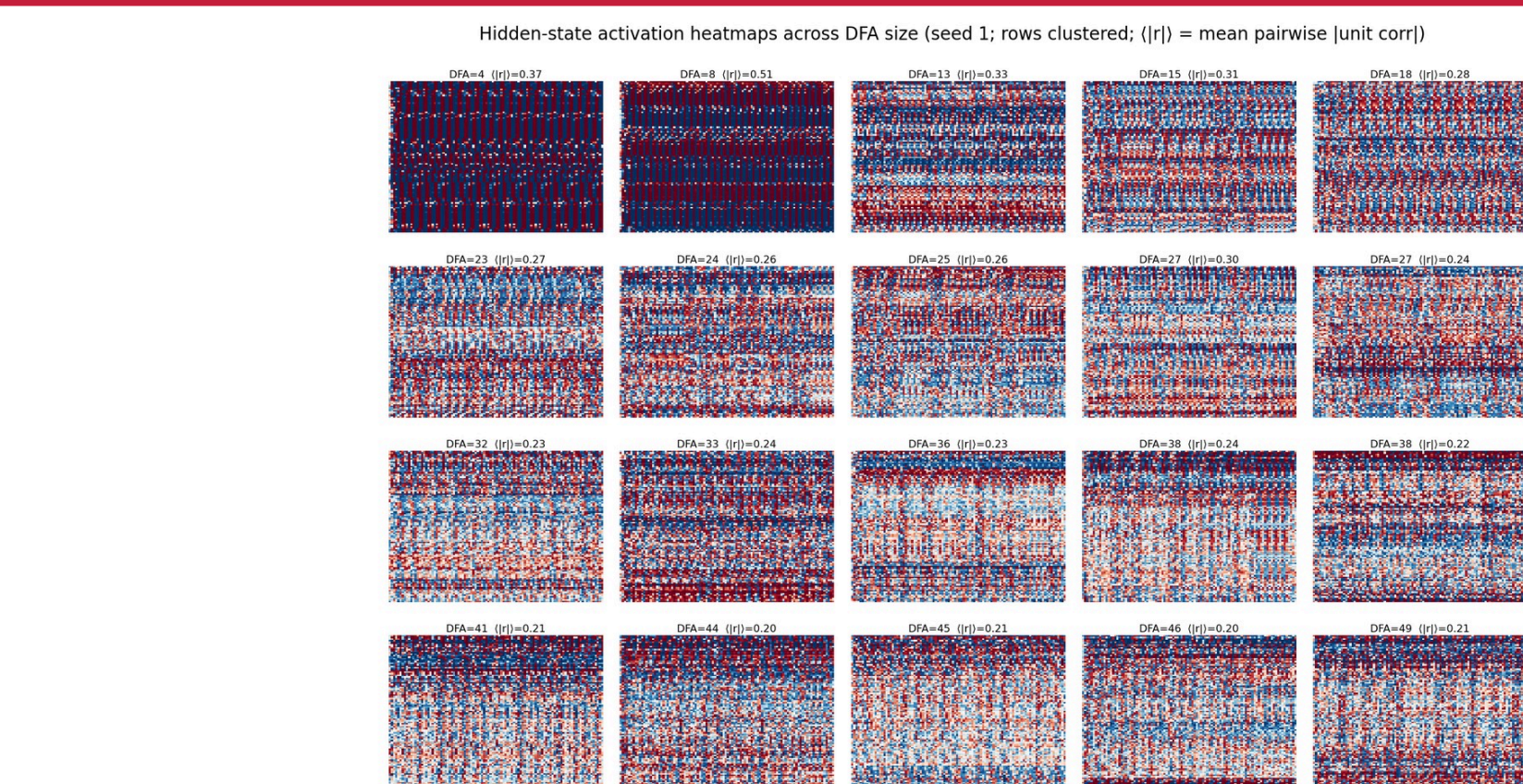
DFA-labeled PCA (top) and closed-loop trajectories (bottom) across five seeds.

Fig 12 • Mixed vocab track DFA size



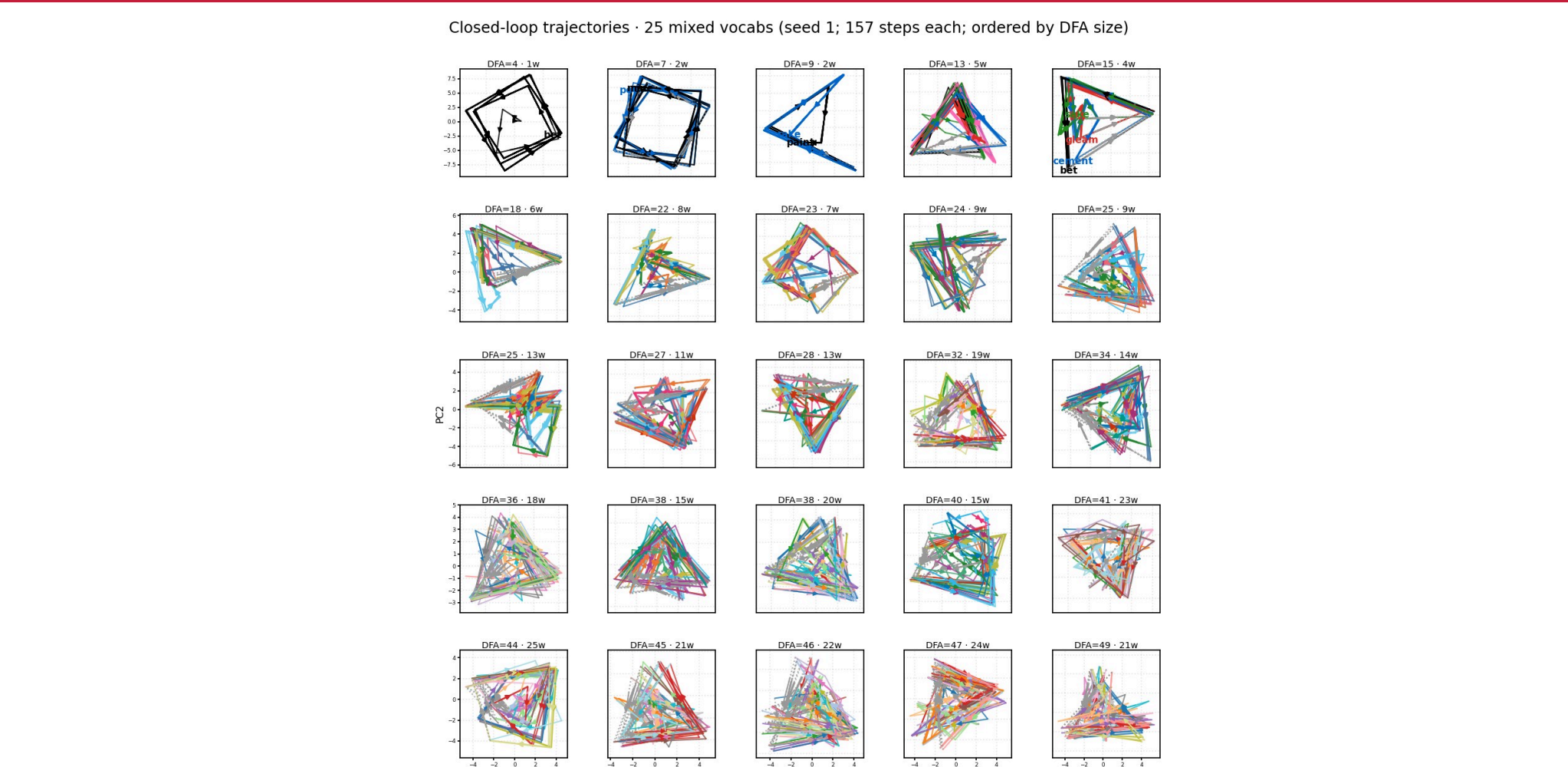
50 mixed English vocabularies ($H=100$). Training cost and PC spectra track DFA size.

Fig 13 • Activations vs DFA size



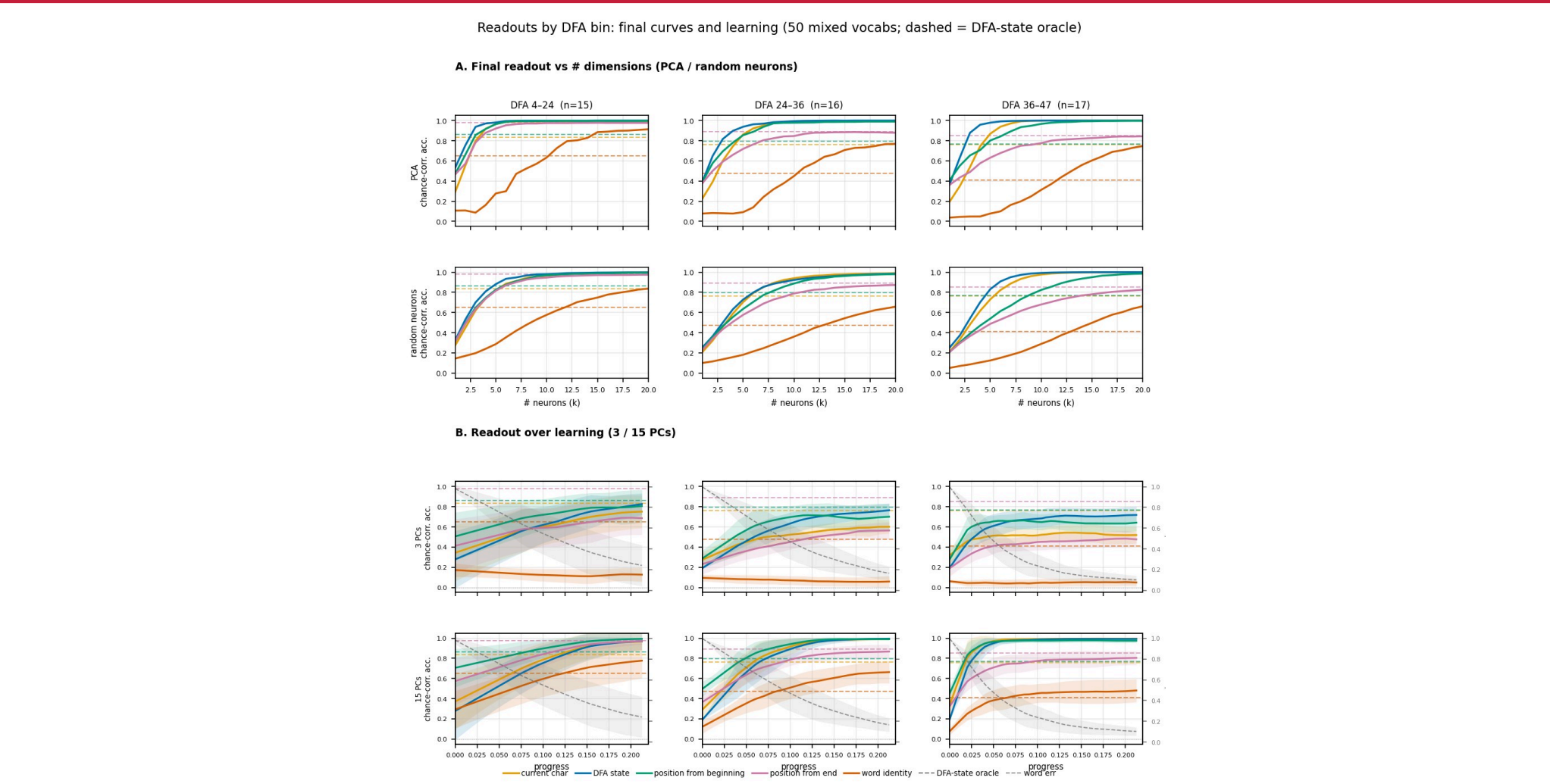
20 mixed vocabularies. Low-DFA: coarse bands; larger DFAs: denser, less repetitive raters.

Fig 14 • Trajectories vs DFA size



25 mixed vocabularies. Small automata: sparse motifs; larger DFAs crowd the PC plane.

Fig 15 • Readouts vs DFA size



Chance-corrected readout by DFA-size bin. Word identity needs ~ 15 PCs.

Fig 16 • Geometry & unit selectivity

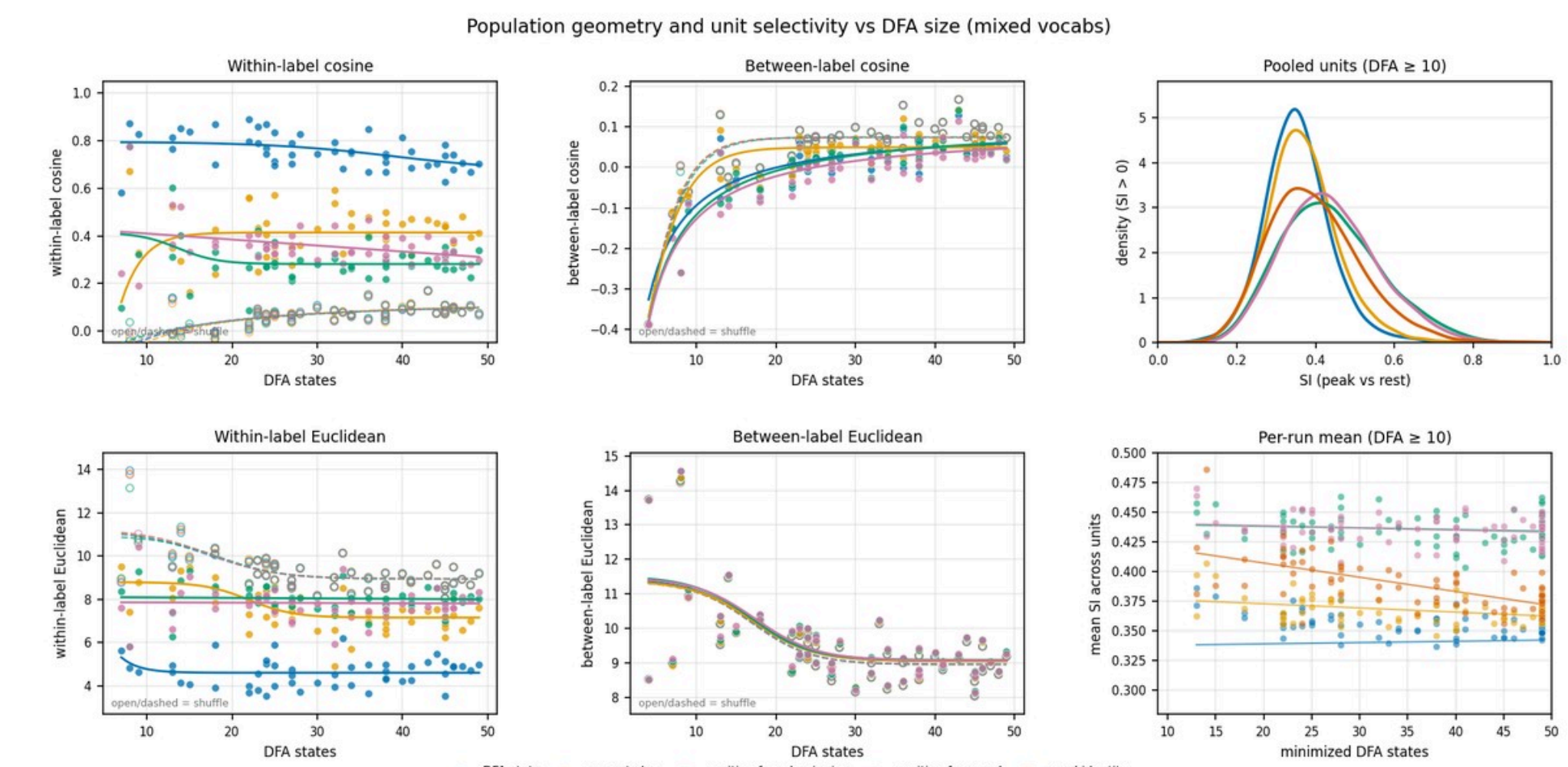
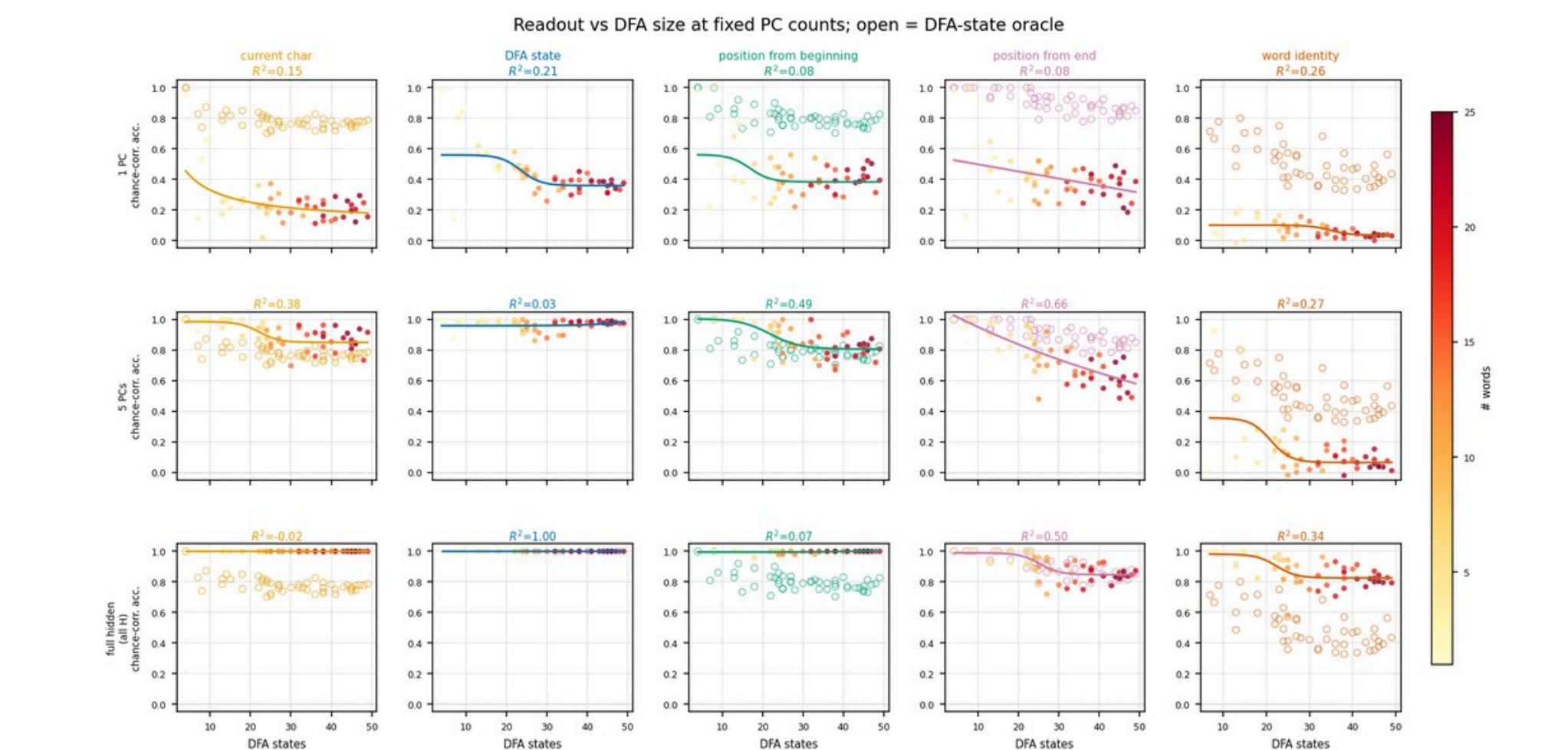
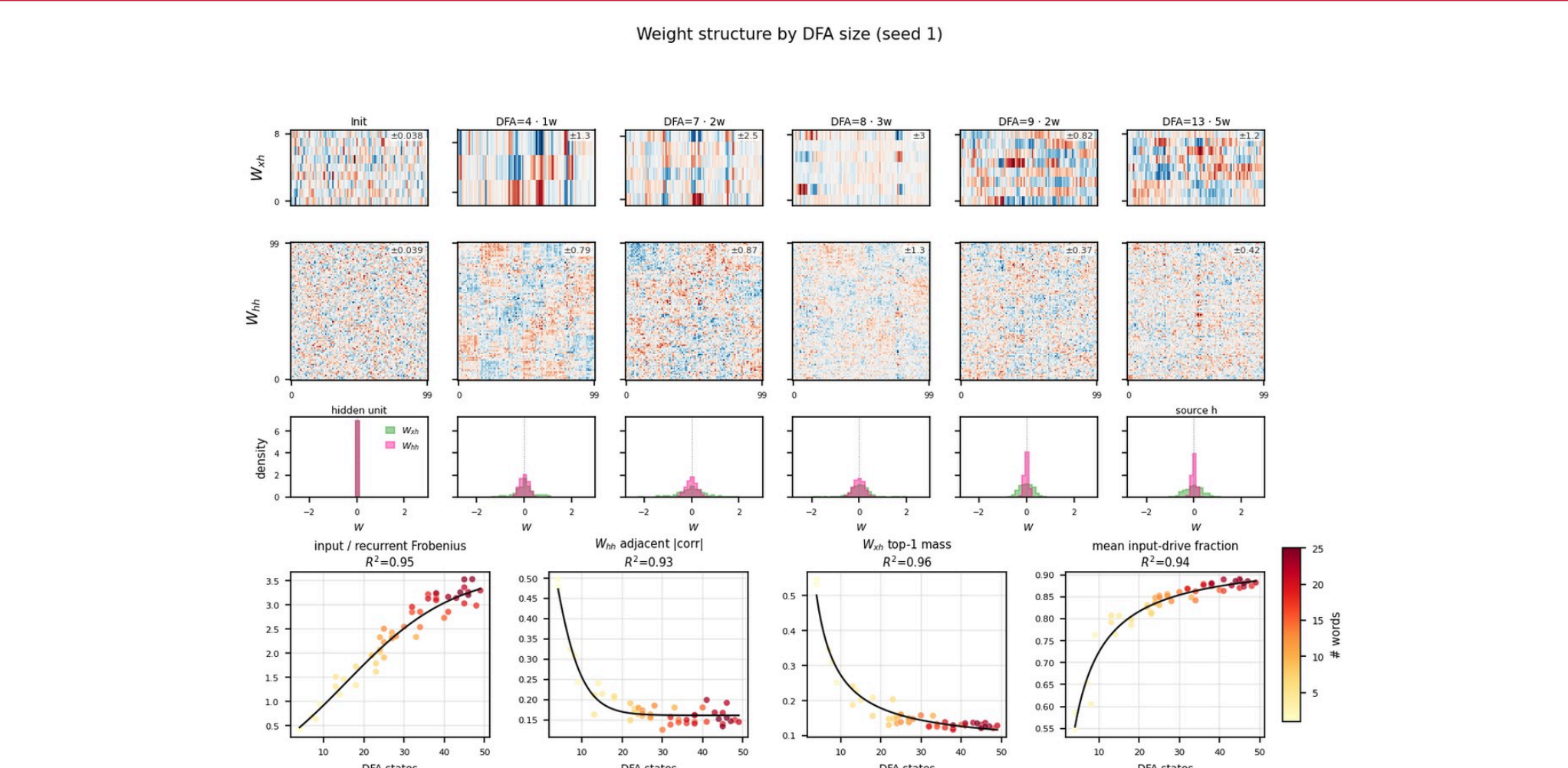


Fig 17 • Readout vs DFA at fixed PCs



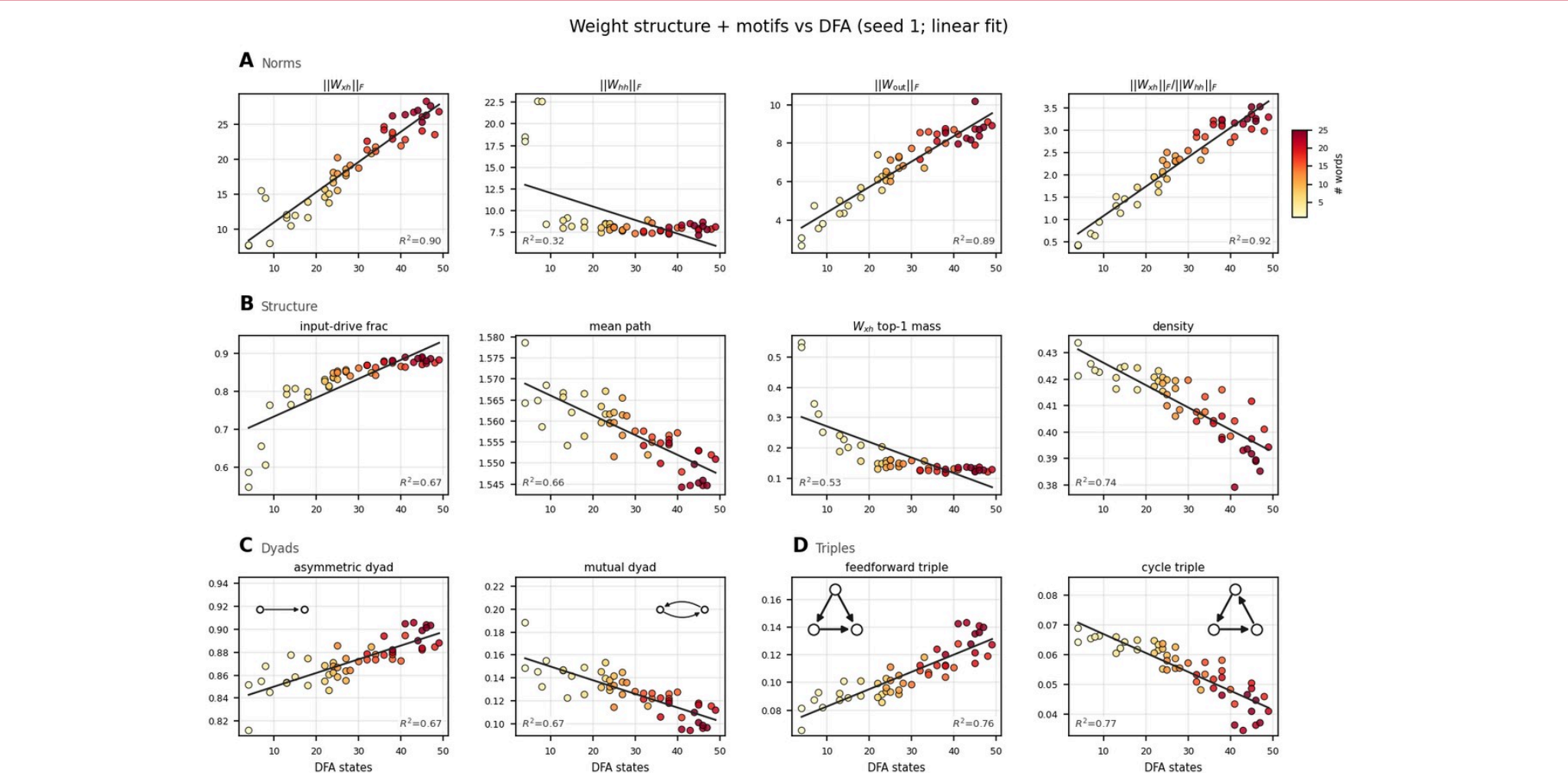
Accuracy vs DFA size at 1 PC, 5 PCs, and full H . Word identity falls off with DFA size.

Fig 18 • Weight matrices vs DFA



Clustered W_{ab} / W_{ba} . Small DFAs: letter columns and local W_{ba} clumps.

Fig 19 • Weight graph motifs vs DFA



Norms, structure, pairwise and 3-node motifs. Larger DFAs shift feedforward.